



Advanced Digital Image Processing Lecture #2

Dr. Md. Hasanuzzaman

Assistant Professor

Department of Computer Science & Engineering
Dhaka University

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Introduction to Digital Image Processing

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- ❖ Common Values of Digital Image Parameters
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Definition of Image

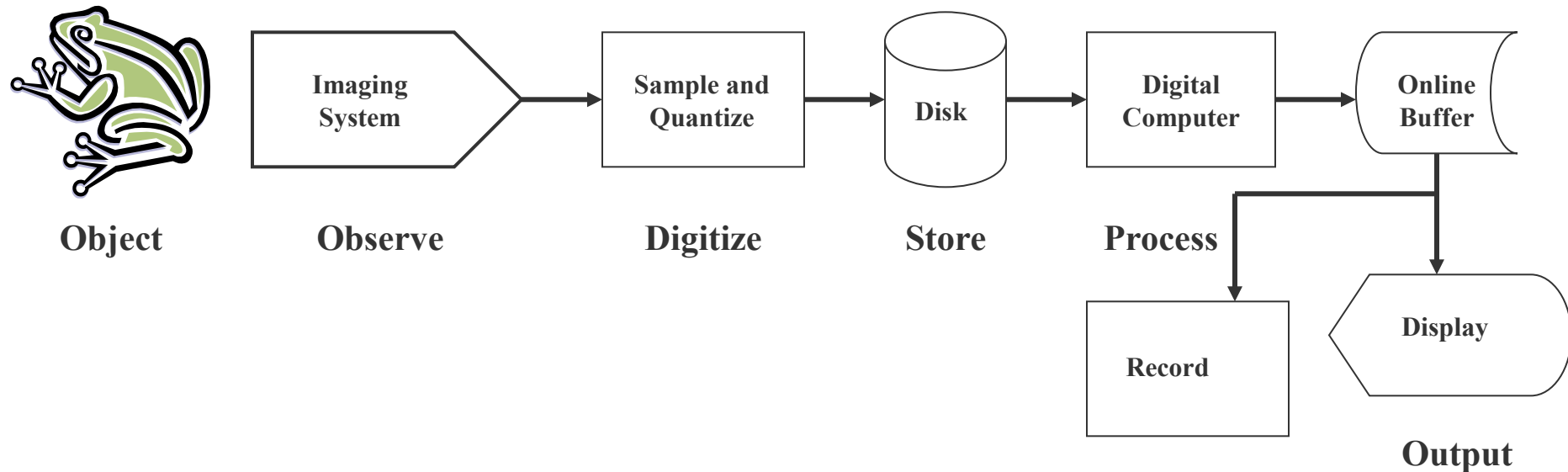
- ❖ An image defined in the "real world" is considered to be a function of two real variables, for example, $a(x,y)$ with a as the amplitude (e.g. brightness) of the image at the *real* coordinate position (x,y) .
- ❖ A digital image $a[m,n]$ described in a 2D discrete space is derived from an analog image $a(x,y)$ in a 2D continuous space through a *sampling* process that is frequently referred to as digitization.



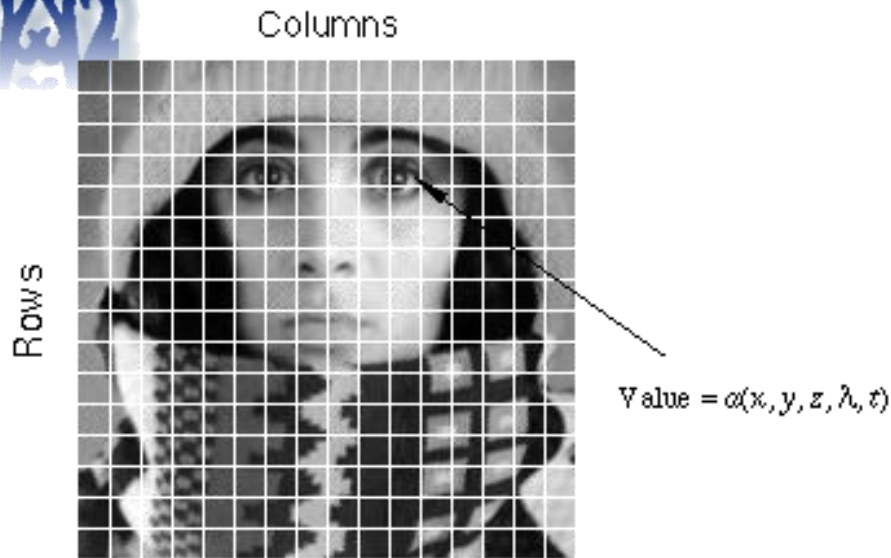
Manipulation on Images

- ❖ Image Processing: *image in* $\rightarrow$ *image out*
- ❖ Image Analysis: *image in* $\rightarrow$ *measurements out*
- ❖ Image Understanding: *image in* $\rightarrow$ *high-level description out*

Block Diagram of a Typical Image Processing Sequence



Example of Digitization



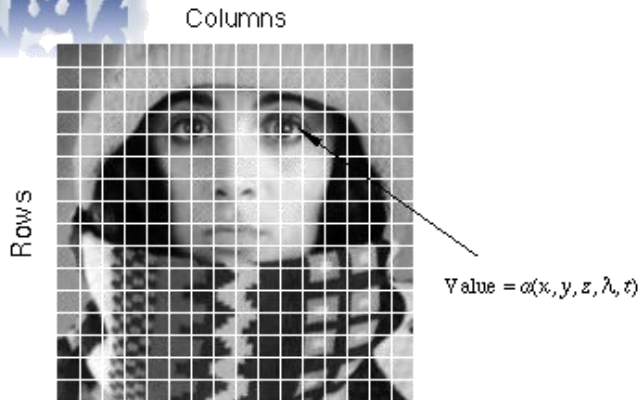
✓ The 2D continuous image $I(x,y)$ is divided into N rows and M columns.

✓ The intersection of a row and a column is termed a **pixel**.

✓ The value assigned to the integer coordinates $[m,n]$ with $\{m=0,1,2,...,M-1\}$ and $\{n=0,1,2,...,N-1\}$ is $I[m,n]$.

depth (z), color (λ), time (t)

Quantization



❖ Image divided into $N = 16$ rows and $M = 16$ columns.

- ❖ The process of representing the amplitude of the 2D signal at a given coordinate as an integer value with L different gray levels is usually referred to as amplitude quantization or simply *quantization*
- ❖ The value assigned to every pixel is the average brightness in the pixel rounded to the nearest integer value.

Common Values of Digital Image Parameters

- *Parameter-> Symbol-> Typical values*
 - Rows-> N -> 256,512,525,625,1024,1035
 - Columns-> M ->256,512,768,1024,1320
 - Gray Levels-> L ->2,64,256,1024,4096,16384
- $M=N=2^K$ where $\{K = 8,9,10\}$.
 - The number of distinct gray levels is usually a power of 2, that is, $L=2^B$ where B is the number of bits in the binary representation of the brightness levels.
 - $B>1$ for a *gray-level image*; $B=1$ for a *binary image*.



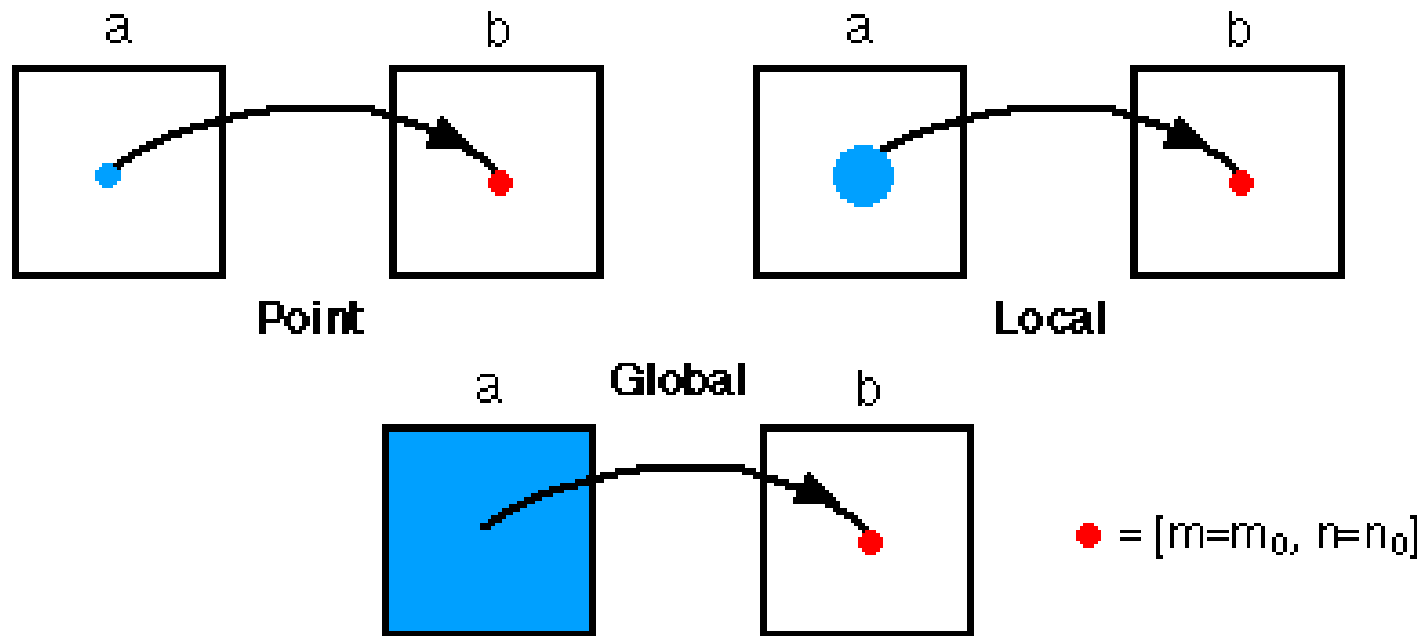
Characteristics of Image Operations

- ❖ Types of Image Operation
- ❖ Types of neighborhoods

Types of Image Operation (1)

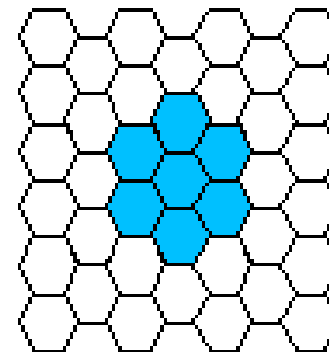
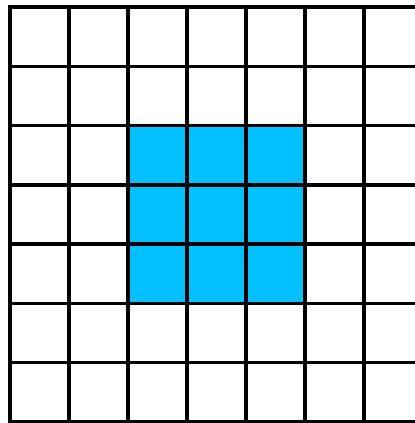
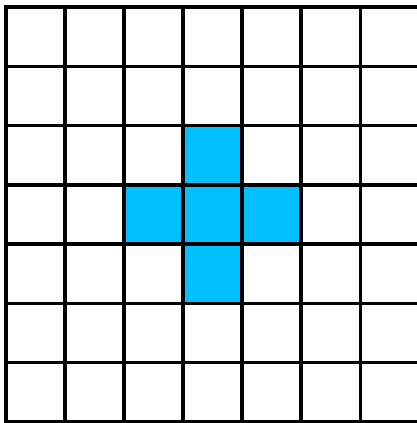
Operation	Characterization	Generic Complexity/Pixel	
* <i>Point</i>	- the output value at a specific coordinate is dependent only on the input value at that same coordinate.	<i>constant</i>	Image size = $N \times N$; neighborhood size = $P \times P$. Note that the complexity is specified in operations <i>per pixel</i>
* <i>Local</i>	- the output value at a specific coordinate is dependent on the input values in the <i>neighborhood</i> of that same coordinate.	P^2	
* <i>Global</i>	- the output value at a specific coordinate is dependent on all the values in the input image.	N^2	

Types of Image Operation (2)



Local operations produce an output pixel value $b[m=m_o, n=n_o]$ based upon the pixel values in the *neighborhood* of $a[m=m_o, n=n_o]$.

Types of Neighborhoods



Rectangular sampling: 4-connected 8-connected

Hexagonal sampling: 6-connected



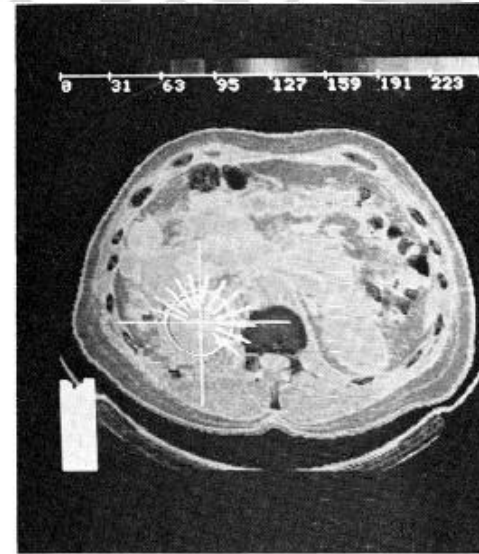
Applications of Digital Image Processing

- ❖ Remote sensing Via Satellites and other Spacecrafts
- ❖ Image Transmission and Storage for Business Applications
- ❖ Medical Image Processing
- ❖ Radar
- ❖ Robotics
- ❖ Automatic Inspection of Industrial Parts
- ❖ Person Identification for Security Measure, etc.

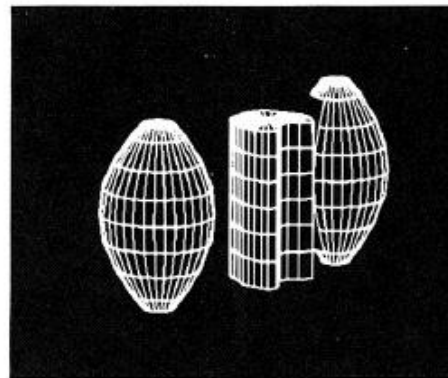
Application: Medical Image



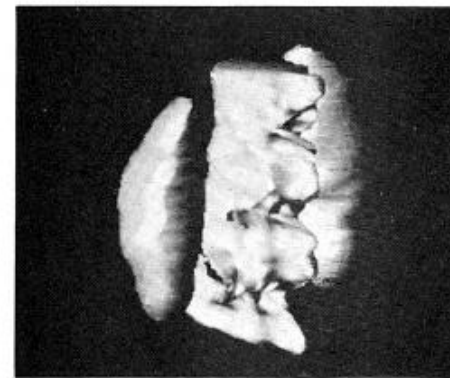
Scan Data^(a)



Model Fitting



Prototype Kidney Model^(b)



Resulting Kidney
and spinal cord in
stance^(d)



Application: Face Detection



Application: Human-Robot Interaction



End of Presentation

Thanks to all !!!